

Types of Scheduling



1. Long-term scheduling:

Decides which processes will be added to the system to run.

2. Medium-term scheduling:

Chooses which processes will stay in memory to continue running.

3. Short-term scheduling:

Picks which process will run next on the processor.

4. I/O scheduling:

Decides which process will use the I/O device next when it is ready.



Processor Scheduling

The goal is to assign processes to the processor in a way that improves system performance, such as faster response, better throughput, and efficient use of the processor.

Processor scheduling is divided into three parts:

1. Long-term scheduling

2. Medium-term scheduling

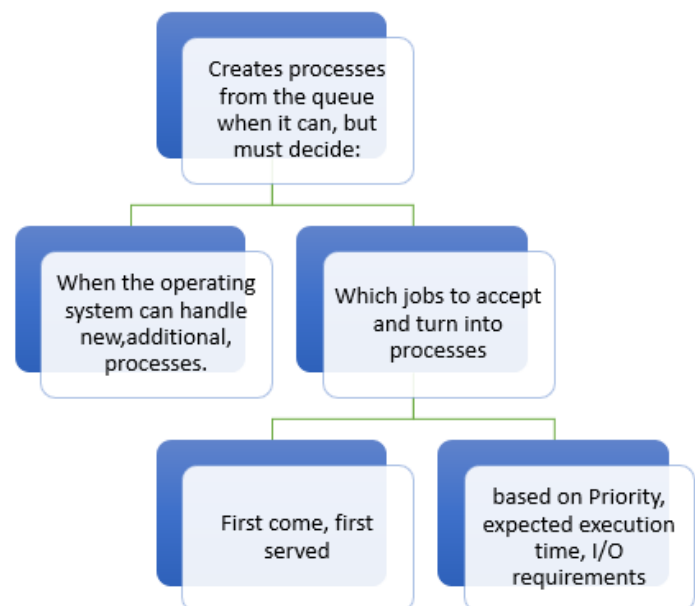
3. Short-term scheduling

Long-Term Scheduler:

- Decides which programs can enter the system for processing.
- Controls how many programs run at the same time (degree of multiprogramming).
 - If too many processes are created, each process will get **less CPU time**.
 - It may limit the number of processes to ensure good performance.

The scheduler must decide:

- When the operating system can handle new, additional, processes.
- Which jobs to accept and turn into processes.
- How to select jobs (e.g., first come, first served or based on priority).



Medium-Term Scheduling

- Manages the swapping function (**moving processes between memory and disk**). !
- Decides when to bring swapped-out processes back into memory. with swapping , it managing degree of multiprogramming.
- These decisions depend on how many processes the system can handle and the memory needed by the processes. !

Short-Term Scheduling

- Also called the **dispatcher**. (Dispatcher selects which process will use the CPU and starts it. it quickly switches between processes.)
- Executes most frequently.(Dispatcher works a lot.it Bc. it quickly switches between processes.
- Makes the **fine-grained decision** of which process to execute next.(Decides which process will use the CPU.)

Examples:

- Clock interrupts
- I/O interrupts
- Operating system calls
- Signals like semaphores



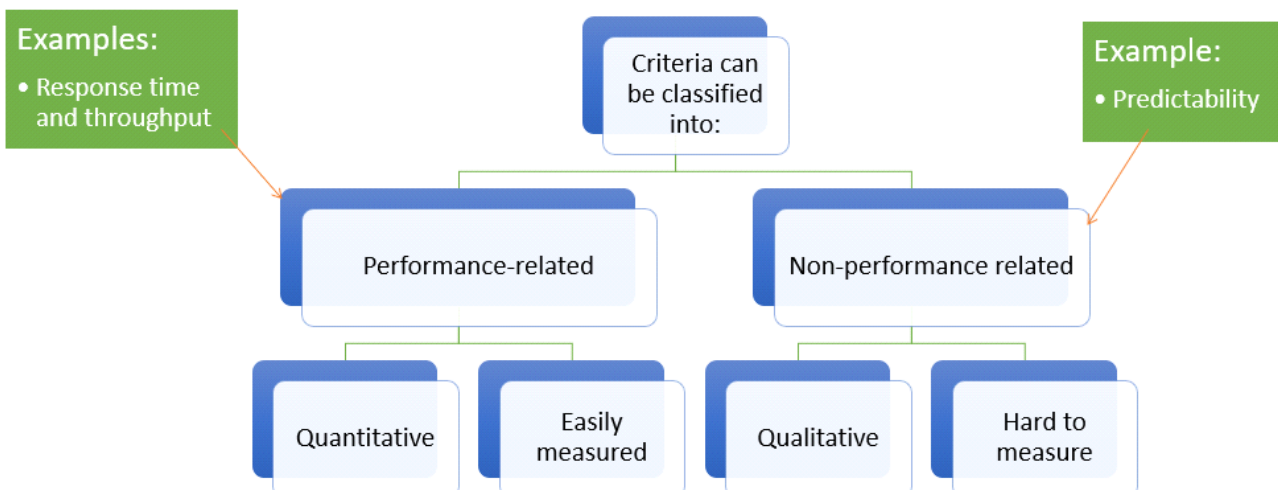
Short-Term Scheduling Criteria

The main goal is to share processor time to make the system work better. To do this, specific criteria are used to evaluate the scheduling.



- **User-oriented criteria**
 - Focuses on how the user or process experiences the system (e.g., fast response time).
 - Important for most systems.
- **System-oriented criteria**
 - Focuses on how well the processor is used (e.g., how many processes are completed).
 - Less important in single-user systems.

Short-Term Scheduling Criteria: Performance



Selection Function !!!!!

- Decides which process, from the ready processes, will run next.
- The decision can be based on:
 - **Priority**
 - **Resource needs**
 - **Process execution characteristics**

If the decision is based on execution characteristics, important values are:



- **w**: Time the process has been waiting.
- **e**: Time the process has been running.
- **s**: Total time needed to finish the process (includes **e** and usually needs to be estimated).

Decision Mode

It shows when the selection function chooses the next process to run.

Two categories: 1. Nonpreemptive 2. Preemptive



Nonpreemptive vs Preemptive

1. Nonpreemptive:

- Once a process starts running, it continues until it finishes or waits for I/O.

2. Preemptive:

- The operating system can stop a running process and move it back to the ready state.
- This can happen when a new process arrives, an interrupt occurs, or based on a timer (clock interrupt).



First-Come-First-Served (FCFS)

- It is the simplest scheduling method.
- Also called **FIFO** (First-In-First-Out) because processes are handled in the order they arrive.
- When a process is ready, it goes into the queue.
- Works better for long processes than short ones.
- Prefers CPU-bound processes over I/O-bound processes.



Round Robin

- Uses **preemption** based on a clock.
- Also called **time slicing** because each process gets a slice of time to run before being preempted.
- Principal design issue is the **length of the time quantum**, or slice, to be used
- Works well in systems where multiple users share the CPU or in transaction processing systems.
- One problem is that it does not treat CPU-bound and I/O-bound processes equally.

clock (zamanlayıcı)

preemption (öncelikli kesme)



Shortest Process Next (SPN)



- It is a **nonpreemptive** scheduling policy.
- The process with the **shortest expected processing time** is chosen to run next.
- Short processes will jump to the front of the queue.
- One problem is that **longer processes may starve** (they may never get a chance to run).
- It is **hard to estimate process time**.
- If the estimate is much shorter than the real time needed, the system may cancel the process.

Shortest Remaining Time (SRT)

- **Preemptive** version of SPN.
- Chooses the process with the shortest remaining time.
- Longer processes may starve.
- Gives better turnaround time than SPN because short jobs get priority.

Highest Response Ratio Next (HRRN)

- Chooses the process with the highest ratio.
- Considers how long a process has waited, giving older processes a higher chance.
- Shorter jobs are preferred, but longer jobs eventually get priority as their ratio increases.